DNA II

16

Correctly complete the definitions below (1–8) using these words:

bases	*DNA*	*homologues*	*hydrogen*
nucleotide	*polypeptide*	*replication*	*template*

1 A huge molecule made up of repeating units of four nucleotides. _____

2 The production of new strands of DNA. _____

3 A three-part unit made of a nitrogen compound, a sugar and a phosphate. _____

4 An abbreviation for nitrogen-based substances found in DNA. _____

5 The model on which a new strand of DNA is built. _____

6 Corresponding chromosomes in a cell. _____

7 Proteins are made of chains of these. _____

8 Weak bonds holding strands of DNA in a double helix. _____

Quick thinking!

9 You may have seen the symbols ♂ and ♀ used in genetics. What does each of these symbols signify?

♂ means _____ ♀ means _____

10 Research where the symbols referred to in Question 9 originated.

_____

11 One strand of DNA replicates a total of six times. How many strands will result?

_____

Alleles

The word 'allele' comes from the Greek, meaning 'of one form and another'. Alleles are the alternate forms of a gene. They are genes occupying the same position on homologous chromosomes. For example eye colour is determined by a gene, but that gene has several different alleles: brown, blue, green, grey and, occasionally, violet. If you consider genes as cake mixes on sale in a supermarket, the alleles would be the different flavours of cake mix available. All the mixes (alleles) would produce a cake, but the one you buy and take home, will determine what eventually comes out of the oven. In a similar manner, you will inherit an eye colour allele from each of your parents and these will pair up to form an eye colour gene. The two alleles you inherit will determine your eye colour.

Consider a child whose mother has blue eyes and whose father has brown eyes.

12 Will the child have one brown eye and one blue eye? _____

13 Will the child have an eye colour that is a blend of brown and blue? _____

14 What eye colour might the child have? Explain your answer. _____

_____

15 Describe another characteristic that is inherited from parents on an 'either/or' basis. _____

16 Describe a characteristic that is inherited from parents on a 'blend/mixture' basis. _____

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